ic_autolab round 1 (generic track): report

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Protocol: PROTOCOL.md, frozen 2026-11-04T01:43:53Z at git `5840364fa4f38a0c47dcdd7f167c580394813cb1`; route `local`. Counted GAE only (deterministic, host independent); no wall-time claim; development loop, no validation-gate row. Loop home: `../`; ledger: `../ledger.json`.

Outcome

k	class	sel	sel / inc at w = 0 [95%]	w = 0.1	w = 1	band	S sel (w = 0 / 0.1 / 1)	S inc	S rho-neg	distance sel / rho
20	**no_challenger**	none (best candidate `ksum3-n15`, train S 10.552 / 10.993 / 14.965)	-	-	-	-	-	8.998 / 9.734 / 16.356	1.501 / 1.607 / 2.560	inc: 6.0x / 6.1x / 6.4x
24	**no_challenger**	none (best candidate `ksum3-n14`, train S 10.355 / 11.908 / 25.885)	-	-	-	-	-	8.053 / 8.760 / 15.129	1.372 / 1.497 / 2.625	inc: 5.9x / 5.9x / 5.8x

Classes follow the protocol's decision rule. Distance to rho is the distance remaining, not a gate. The reasons per size:

- k = 20: no_challenger
- k = 24: no_challenger

Train (development screen, in sample)

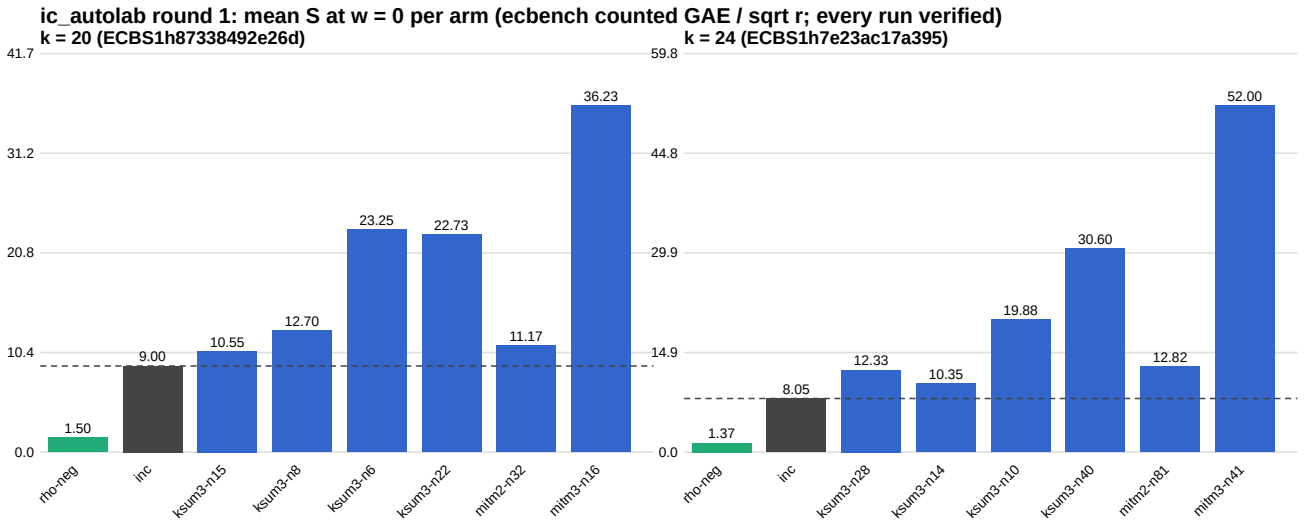
k = 20: session `ECBS1h87338492e26d` (run exit 0, verify exit 0, receipt sha256 `5390d9bcbc9286a55d95b58c88c89813625f0a1b4699b1ac5ef90bd2509ac75b`)

arm	family	N	factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`rho-neg`	rho	-	-	`rho.negation`	16	16	1.501 / 1.607 / 2.560	matched reference
`inc`	ksum3	11	`prime-projected-abscissa:size=11,scan_cap=176`	`ksum:m=3`	16	16	8.998 / 9.734 / 16.356	incumbent
`ksum3-n15`	ksum3	15	`prime-projected-abscissa:size=15,scan_cap=240`	`ksum:m=3`	16	16	10.552 / 10.993 / 14.965	incumbent neighbourhood: N = 1.4 x 11
`ksum3-n8`	ksum3	8	`prime-projected-abscissa:size=8,scan_cap=128`	`ksum:m=3`	16	16	12.697 / 14.226 / 27.988	incumbent neighbourhood: N = 0.7 x 11
`ksum3-n6`	ksum3	6	`prime-projected-abscissa:size=6,scan_cap=96`	`ksum:m=3`	16	16	23.251 / 26.894 / 59.674	incumbent neighbourhood: N = 0.5 x 11
`ksum3-n22`	ksum3	22	`prime-projected-abscissa:size=22,scan_cap=352`	`ksum:m=3`	16	16	22.729 / 23.087 / 26.301	incumbent neighbourhood: N = 2 x 11
`mitm2-n32`	mitm2	32	`prime-projected-abscissa:size=32,scan_cap=512`	`mitm:negation_folded=1`	16	16	11.170 / 11.972 / 19.189	exploration: family mitm2 untried at k = 20, N = 1 x N_a

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`mitm3-n16`	mitm3	16 `prime-projected-abscissa:size=16,scan_cap=256`	`mitm:m=3,negation_folded=1`	16	16	36.227 / 39.560 / 69.555	exploration: family mitm3 untried at k = 20, N = 0.5 x N_a

k = 24: session `ECBS1h7e23ac17a395` (run exit 0, verify exit 0, receipt sha256 `95f7bc1fe91f99a3ebe501b10a3ccb06b3b074a96c3101d1d41c820e465a926f`)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`rho-neg`	rho	- -	`rho.negation`	16	16	1.372 / 1.497 / 2.625	matched reference
`inc`	ksum3	20 `prime-projected-abscissa:size=20,scan_cap=320`	`ksum:m=3`	16	16	8.053 / 8.760 / 15.129	incumbent
`ksum3-n28`	ksum3	28 `prime-projected-abscissa:size=28,scan_cap=448`	`ksum:m=3`	16	16	12.330 / 12.699 / 16.021	incumbent neighbourhood: N = 1.4 x 20
`ksum3-n14`	ksum3	14 `prime-projected-abscissa:size=14,scan_cap=224`	`ksum:m=3`	16	16	10.355 / 11.908 / 25.885	incumbent neighbourhood: N = 0.7 x 20
`ksum3-n10`	ksum3	10 `prime-projected-abscissa:size=10,scan_cap=160`	`ksum:m=3`	16	16	19.881 / 23.476 / 55.830	incumbent neighbourhood: N = 0.5 x 20
`ksum3-n40`	ksum3	40 `prime-projected-abscissa:size=40,scan_cap=640`	`ksum:m=3`	16	16	30.597 / 30.879 / 33.419	incumbent neighbourhood: N = 2 x 20
`mitm2-n81`	mitm2	81 `prime-projected-abscissa:size=81,scan_cap=1296`	`mitm:negation_folded=1`	16	16	12.816 / 13.989 / 24.547	exploration: family mitm2 untried at k = 24, N = 1 x N_a
`mitm3-n41`	mitm3	41 `prime-projected-abscissa:size=41,scan_cap=656`	`mitm:m=3,negation_folded=1`	16	16	51.995 / 57.066 / 102.697	exploration: family mitm3 untried at k = 24, N = 0.5 x N_a



grey: incumbent; green: matched rho; blue: candidate (every run verified); red: candidate with an unverified or failed run. Bars are recorded band means; the dashed line is the incumbent.

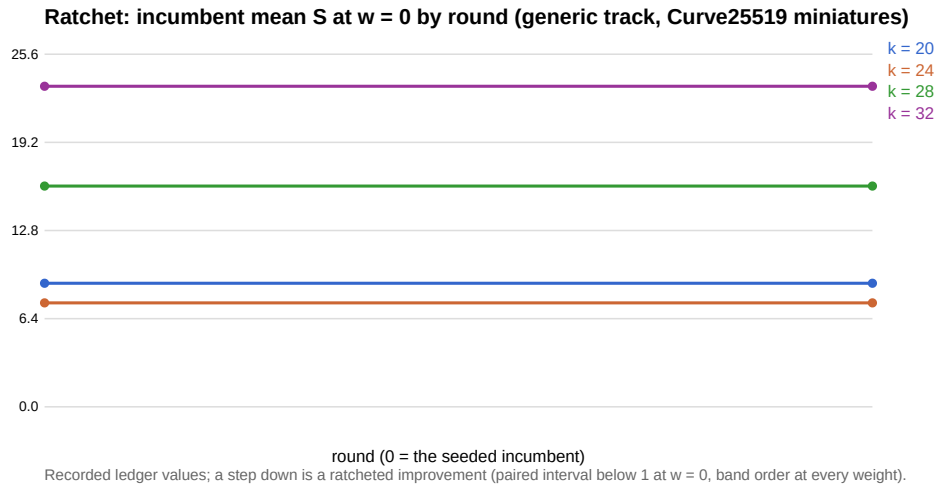
train screen

Selection

- k = 20: no_challenger (Pareto alternative at w = 1: `ksum3-n15`)
- k = 24: no_challenger (Pareto alternative at w = 1: `ksum3-n28`)

Ledger after this round

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
20	`ksum3-n11` (`prime-projected-abscissa:size=11,scan_cap=176` + `ksum:m=3`)	0	8.987 / 9.710 / 16.217	5.7x / 5.7x / 6.0x
24	`ksum3-n20` (`prime-projected-abscissa:size=20,scan_cap=320` + `ksum:m=3`)	0	7.554 / 8.173 / 13.745	6.5x / 6.4x / 6.2x
28	`mitm2-n203` (`prime-projected-abscissa:size=203,scan_cap=3248` + `mitm:negation_folded=1`)	0	16.057 / 17.564 / 31.130	16.6x / 16.6x / 16.7x
32	`mitm2-n512` (`prime-projected-abscissa:size=512,scan_cap=8192` + `mitm:negation_folded=1`)	0	23.310 / 25.520 / 45.410	24.7x / 24.6x / 24.3x



ratchet

Evidence and deviations

- ecbench binary sha256: `3fd22853485e388782ac0821295c56f81725fc832fd3610978be8088b5098bca`; `--cpus auto`, allow_busy = true; `verify --replay 8` per session; audit receipts in `analysis/`; sessions in `sessions/` (records in the artifact store when large).
- Route: the local route is an explicitly requested development screen, not the Iso Lab EKS default; counts are deterministic and replayed, wall time is not claimed. No row of this round enters the IC leaderboard or `benchmarks/README.md` (validation gate).
- Train selection is in sample (disclosed); the holdout is fresh (new `target_seed`, excluded exposures). Failed, timed-out and regressing arms are retained above.